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Supplemental inflatable restraint installation system

Abstract

A supplemental inflatable restraint installation system for roof rail airbags in vehicles. A supplemental restraint system for a vehicle includes two mounts connected to the vehicle. An inflatable bag includes a retainer, and a rod extends between the two mounts. The rod extends through the retainer to secure the inflatable bag to the vehicle with a minimum number of fastening points.

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Background/Summary

INTRODUCTION

(1) The technical field generally relates to supplemental inflatable restraint systems in vehicles and installation systems thereof, and more particularly relates to roof rail or curtain airbags that include structural members configured to install the airbags in a vehicle with a minimal number of fasteners.

(2) Supplemental inflatable restraint systems of the airbag type, including those referred to as curtain airbags or side curtain airbags, are features in vehicles designed to deploy and restrain occupants under certain conditions. Unlike supplemental inflatable restraints that deploy from the steering wheel or dashboard, roof rail airbags deploy from the trim components above or around the side windows, such as from under the headliner, and when deployed may cover the window areas like curtains.

(3) In certain events, sensors in the vehicle detect the conditions and trigger the deployment of the roof rail airbags. The airbags inflate rapidly, forming a barrier along the side windows and pillars. The inflated airbags may cover the surfaces of the vehicle's interior, such as the side windows and/or pillars. Roof rail airbags may extend along any part of the cabin's side along the front to the rear of the vehicle's cabin, covering the side window area.
(4) To maintain an airbag in its desired position, a number of attachment points may be used along the roof rails and/or pillars of the vehicle body. Installation may require numerous fastening operations to provide the desired amount of retention. Accordingly, there is an ongoing desire for systems that simplify assemble operations and to deliver desired functionality of roof rail airbags. Furthermore, other desirable features and characteristics of the present disclosure will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and the foregoing introduction.

SUMMARY

(5) A supplemental inflatable restraint installation system is provided for vehicles. In a number of embodiments, a supplemental restraint system for a vehicle includes a pair of fasteners connected to the vehicle. An inflatable bag has a loop. A rod extends between, and is connected with, the fasteners. The rod extends through the loop and retains the inflatable bag to the vehicle.
(6) In additional embodiments, the rod has a slip fit within the retainer.
(7) In additional embodiments, the rod is conformable to a profile of the vehicle.
(8) In additional embodiments, the vehicle includes a body with a roof rail, and the inflatable bag is secured to the roof rail by the mounts and the rod.
(9) In additional embodiments, the airbag and the retainer are fabricated from a common material.
(10) In additional embodiments, the retainer has a sleeve through which the rod extends. The sleeve extends substantially completely between the mounts.
(11) In additional embodiments, the mounts each have a structure that has a barrel section and a strap section, and the rod extends through the barrel section of the mounts.
(12) In additional embodiments, the mounts each have a structure that has a barrel section and a strap section. The rod extends through the barrel section of each of the mounts, and the strap section of each of the mounts is secured to the vehicle.
(13) In additional embodiments, the retainer completely encircles the rod.
(14) In additional embodiments, the rod is secured to the mounts in fixed positions.
(15) In a number of additional embodiments, a supplemental restraint system for a vehicle includes a pair of mounts fixed to the vehicle and includes an inflatable bag. A retainer is connected with the inflatable bag, and a rod extends between, and is connected with, the mounts. The rod extends through the retainer to secure the inflatable bag to the vehicle.
(16) In additional embodiments, the retainer operates to slip along the rod while remaining secured to the rod during a deployment of the inflatable bag.
(17) In additional embodiments, the rod is conformable to a profile of the vehicle during installation of the inflatable bag in the vehicle.
(18) In additional embodiments, the vehicle includes a body with a roof rail and an A-pillar, and the inflatable bag is secured to the roof rail and the A-pillar.
(19) In additional embodiments, the airbag and the retainer are fabricated from a common woven material.
(20) In additional embodiments, the retainer has a sleeve through which the rod extends, and the sleeve extends substantially completely between the mounts with a gap in the sleeve at one of the mounts.
(21) In additional embodiments, the mounts each have a structure that has a barrel section and a strap section integral with the barrel section, and the rod extends through the barrel section of each of the mounts.
(22) In additional embodiments, the mounts each have a structure that has a barrel section and a

strap section integral with the barrel section. The rod extends through the barrel section of each of the mounts. The strap section of each of the mounts is secured to the vehicle by a fastener.
(23) In additional embodiments, the rod extends from one mount on both of its sides, and the inflatable bag is connected with the rod on both sides.
(24) In a number of other embodiments, a supplemental restraint system for a vehicle includes a body of the vehicle that has a roof rail. A pair of mounts are fixed to the roof rail. An inflatable bag is included. A retainer is connected with the inflatable bag and has a structure that loops over the rod. The structure includes ends that are each fixed to the inflatable bag. A rod extends between, and is connected with, the mounts. The rod extends through the retainer to secure the inflatable bag to the vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The exemplary embodiments will hereinafter be described in conjunction with the following drawing figures, wherein like numerals denote like elements, and wherein:
(2) FIG. **1** is a functional block diagram of a vehicle including a supplemental inflatable restraint installation system for roof rail type airbags, in accordance with various embodiments;
(3) FIG. **2** is an interior view of the roof airbag installed to a roof rail of the vehicle of FIG. **1** in accordance with various embodiments;
(4) FIG. **3** is a perspective view of a portion of the roof airbag of FIG. **2**, in accordance with various embodiments;
(5) FIG. **4** is a view of the attachment area of the roof airbag of FIGS. **2** and **3**, in accordance with various embodiments; and
(6) FIG. **5** is a sectional view take generally through the line **5-5** in FIG. **4**, in accordance with various embodiments.
DETAILED DESCRIPTION
(7) The following detailed description is merely exemplary in nature and is not intended to limit the application and uses. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding introduction or the following detailed description.
(8) FIG. **1** illustrates a vehicle **10**, according to an embodiment. In certain examples, the vehicle **10** comprises an automobile. The vehicle **10** may include various restraint systems such as a roof airbag system having curtain or roof rail airbags for restraining occupants of the vehicle **10** in certain triggering events. The roof rail airbags may include one or more features configured to promote proper installation and operation of the roof rail airbags, including a type of rod-like structural member that simplifies installation of the airbag system into the vehicle **10**.
(9) In various examples, the vehicle **10** may be any one of a number of different types of automobiles, such as, for example, a sedan, a wagon, a truck, or a sport utility vehicle (SUV), and may be two-wheel drive (2WD) (i.e., rear-wheel drive or front-wheel drive), four-wheel drive (4WD) or all-wheel drive (AWD), and/or various other types of vehicles or mobile platforms in certain examples.
(10) As depicted in FIG. **1**, the exemplary vehicle **10** generally includes a chassis **12**, a body **14**, front wheels **16**, and rear wheels **18**. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a body **15** (FIG. **2**). The wheels **16-18** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**.
(11) The vehicle **10** further includes a propulsion system **20**, a transmission system **22**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, at least one controller **34**, and a supplemental inflatable restraint installation system **36** including or associated with one or more

airbags **35**. The airbag(s) **35** may be a side curtain airbag, and/or a roof rail air bag, or another type. The propulsion system **20** includes an engine and/or motor **21** such as an internal combustion engine (e.g., a gasoline or diesel fueled combustion engine), an electric motor (e.g., a 3-phase AC motor), or a hybrid system that includes more than one type of engine and/or motor. The transmission system **22** is configured to transmit power from the propulsion system **20** to the wheels **16**, **18** according to selectable speed ratios. According to various examples, the transmission system **22** may include a multi-ratio automatic transmission, a continuously variable transmission, or other appropriate transmission. The steering system **24** influences positions of the wheels **16**, **18**.

(12) The sensor system **28** includes one or more sensing devices **40***a*-**40***n* that sense observable conditions of the exterior environment, the interior environment, and/or a status or condition of a corresponding component of the vehicle **10** and provide such condition and/or status to other systems of the vehicle **10**, such as the controller **34**. It should be understood that the vehicle **10** may include any number of the sensing devices **40***a*-**40***n*. The sensing devices **40***a*-**40***n* can include, but are not limited to, current sensors, voltage sensors, temperature sensors, motor speed sensors, position sensors, radars, lidars, global positioning systems, optical cameras, thermal cameras, ultrasonic sensors, inertial measurement units, pressure sensors, position sensors, speed sensors, and/or other sensors. One of more of the sensing devices **40***a*-**40***n* may be functional, individually or in combination, to indicate to the controller **34** that the vehicle **10** has been or will be in a collision or rollover event.

(13) The actuator system **30** includes one or more actuator devices **42***a*-**42***n* that control one or more vehicle features such as, but not limited to, the propulsion system **20**, the transmission system **22**, and/or the supplemental inflatable restraint installation system **36**, including the roof rail airbags **35**. One or more of the actuator devices **42***a*-**42***n* may be functional, individually or in combination, to initiate deployment of the airbag(s) **35** in response to predetermined conditions, such as detection that the vehicle **10** has been or will be in an airbag inflation triggering event.

(14) In some examples, the airbag(s) **35** may include multiple airbags strategically placed throughout the cabin of the vehicle **10**, including front, side, and roof rail airbags, to provide comprehensive protection. If an airbag inflation triggering event occurs, one or more of the sensing devices **40***a*-**40***n* detect, for example, the rapid deceleration of the vehicle **10** and send signals to the controller **34**. In response to these sensed events, the controller **34** triggers the deployment, such as through an inflator **38**, of one or more of the airbags, such as the airbag(s) **35**.

(15) Referring to FIG. **2**, with continued reference to FIG. **1**, an exemplary installation of an airbag **35** is presented with the supplemental inflatable restraint installation system **36**. It should be appreciated that the airbag **35** will be covered/concealed by interior trim components (not shown) of the vehicle **10**, which are omitted for visibility. The airbag **35** may be secured to a roof rail **39** of the body **15** of the vehicle **10** that extends along the top, for example, of an A-pillar **17** of the body **15**. The airbag **35** may also be secured to the A-pillar **17**. In some embodiments, the airbag(s) **35** may extend over a B-pillar **19**, and towards, over, and/or along a C-pillar (not shown) and/or D-pillar (not shown). FIG. **2** shows a portion of the vehicle **10** as including the supplemental inflatable restraint installation system **36** of the airbag **35** secured to and extending along the A pillar **17** and over the B pillar **19** on the roof rail **39**. In this embodiment, the airbag **35** is secured to the body **15** with the supplemental inflatable restraint installation system **36**.

(16) Referring to FIG. **3**, along with FIGS. **1** and **2**, the airbag **35**, includes a deployable, inflatable body configured to automatically deploy, via the inflator **38**, in response to detection of an airbag inflation triggering event (e.g., as sensed by the sensor system **28**, detected by the controller **34**, and initiated by the actuator system **30**) to cover portions of an interior side of a vehicle **10** upon being deployed. The airbag **35** may be made of a semi-flexible material (e.g., a fabric), and may include reinforcements and/or other components at select locations. The airbag **35** is packed, such as by being rolled or folded, when in the undeployed state shown. At least two mounts **23** connect, through a rod **50** (or rods), the airbag **35** with the body **15** of the vehicle **10** such as at the roof rail

39 and/or the A-pillar **17**.
(17) The rod **50** extends between at least two of the mounts **23** and more than two mounts **23** may be used to connect the rod (with the airbag **35**) to the body **15** if desired. The rod **50** may be made of a solid construction, may be hollow, or may be made of a plural number of materials. In embodiments, multiple rods **50** may be used, with each connected to the body **15** by at least two mounts **23**. The rod(s) **50** may have a round cross section or another shape and are made of a material to provide sufficient rigidity to hold the airbag **35** during deployment to withstand kinematic forces. For example, the rod(s) **50** may be made of a metal, such as steel, a composite, such as a carbon (or other) fiber reinforced polymer, or another material. The rod(s) **50** may be pre-formed with bends, or may be conformable during installation to follow the contour of the body **15** to which it is attached. For example, the rod(s) **50** may have sufficient deformability or flexibility to aid in installation to the contour during installation, while having sufficient rigidity to retain the airbag **35** to the vehicle **10**.
(18) The air bag **35** includes a retainer **52** or multiple retainers **52** of a material through which the rod **50** may be inserted, or in which the rod **50** is disposed. In the current embodiment, the retainer **52** is a sleeve, made of material connected to, or formed as part of, the airbag **35**, and the rod **50** extends through the sleeve. In embodiments, the retainers(s) **52** may be made of the same or a different material than the airbag **35** and in any case may by coupled to, or formed with, the airbag **35** such as by sewing, fastening, weaving, bonding, or by other means. In embodiments, the retainer(s) **52** may be straps, hoops, hooks, hangers, braided strands, bands, tabs, rings, loops, sleeves, pockets, tubes, or other configurations that may encircle (at least partially), the rod **50** and function to connect the airbag **35** to the rod **50**. In general, the rod **50** slip fits within the retainer(s) **52**, that may be provided as a part of the assembly of the airbag **35**. Slip fit means that the rod **50** may be inserted through the retainer(s) **52** with no or insignificant resistance. There may be one or a plural number of retainers **52** between each pair of adjacent mounts **23**. During assembly, the rod **50** is inserted through the retainer(s) **52** and the rod **50** is connected to the roof rail **39** by the mounts **23** securing the airbag **35** to the body **15**. Including the rod(s) **50** reduces the number of mounts **23** that would otherwise be needed and simplifies assembly. The retainer(s) **52** may loop over the rod **50** and may include ends of the looping structure that are each fixed to the inflatable part of airbag **35**. The airbag **35** may include reinforcements at and/or around the retainer(s) **52**.
(19) Referring additionally to FIGS. **4** and **5**, the mounts **23** may be in the form of P-brackets and each includes a hanger **60** and a fastener **62**. The hanger **60** includes a barrel section **64** and a strap section **66**. In the current embodiment, the barrel section **64** and the strap section **66** are integrally formed, such as of a metal. The barrel section **64** includes an opening **68** through which the rod **50** extends and is shaped to encircle the rod **50**. The strap section **66** may be flat and extends from the barrel section **64**. In embodiments, the strap section **66** may be formed with a shape to mate with a contoured part of the body **15**, such as of the roof rail **39**. The fastener **62** connects the strap section **66** to the body **15**. In the current embodiment, the fastener **62** includes a screw **72** with a washer **74**, where the screw **72** extends through the strap section **66** and is threaded into the body **15**. In embodiments, the fastener **62** may be another type of mechanical fastener or the mount **23** may be connected to the body **15** by another means such as welding. The barrel section **64** may be secured to the rod **50** to maintain relative position. For example, the two may be secured by a press fit, a tack weld, a staking operation, bonding, or another means. As a result, when installed in the vehicle **10** the rod **50** may be retained in a fixed position relative to the body **15**.
(20) As shown in FIG. **4**, the airbag **35** includes two retainers **52**, with a gap **76** between the two at the mount **23**. The two retainers **52** are configured as a sleeve to the left (as viewed) of the mount **23** and as a strap to the right (as viewed) of the mount **23**. As shown in FIG. **5**, the retainer **52** is formed of the material of the airbag **35** and completely encircles the rod **50**. The retainer **52** loops over the rod **50** and the looping part/structure includes ends **56**, **58** that are each fixed to/formed with the inflatable part of airbag **35**, Through combination of the airbag **35**, with the retainer(s) **52**,

the rod **50** and the mount(s) **23**, the airbag **35** is securely held to the body **15** and inclusion of the rod **50** provides retention at spaces between adjacent mounts **23**. The retainer(s) **52** may be continuous between adjacent mounts **23**, or substantially continuous between adjacent mounts **23**, to effectively retain the airbag **35** during deployment.

(21) The systems disclosed herein provide various benefits. For example, the systems disclosed herein provide a simplified installation system for airbags such as roof rail airbags with design simplicity and reduced components. In addition, the systems maintain the proper position of the airbag along its attached body components.

(22) While at least one exemplary embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A supplemental inflatable restraint system for a vehicle comprising: a first mount having a first barrel section that has first tapered ends and having a first strap section that is connected to the vehicle; a second mount having a second barrel section that has second tapered ends and having a second strap section that is connected to the vehicle; an inflatable bag having a retainer; and a rod extending between, and connected with, the first mount and the second mount, the rod extending through the retainer and configured to secure the inflatable bag to the vehicle, the first barrel section of the first mount and the second barrel section of the second mount each immovably fixed to the rod to maintain relative position, wherein the first tapered ends and the second tapered ends are disposed along and taper to the rod, wherein the first and second mounts are configured to retain the rod in a fixed position relative to the vehicle.
2. The supplemental inflatable restraint system of claim 1, wherein during assembly the rod is configured to slip fit within the retainer.
3. The supplemental inflatable restraint system of claim 1, wherein the rod is substantially rigid and conformable to a profile of the vehicle during installation.
4. The supplemental inflatable restraint system of claim 1, wherein the vehicle includes a body with a roof rail, wherein the inflatable bag is secured to the roof rail by the mounts and the rod, wherein the rod is rigid and pre-formed before installation with bends that match a profile of the roof rail.
5. The supplemental inflatable restraint system of claim 1, wherein the rod is one of a plural number of rods retaining the inflatable bag, wherein each of the rods is fixed to the vehicle by a pair of mounts.
6. The supplemental inflatable restraint system of claim 1, wherein the retainer comprises a sleeve through which the rod extends, wherein the sleeve extends substantially completely between the first mount and the second mount.
7. The supplemental inflatable restraint system of claim 1, wherein the rod extends through the barrel section of the first and second mounts.
8. The supplemental inflatable restraint system of claim 1, wherein the rod extends through the barrel section of each of the first and second mounts, and wherein the barrel sections are welded to the rod.
9. The supplemental inflatable restraint system of claim 1, wherein the retainer completely encircles the rod.

10. The supplemental inflatable restraint system of claim 1, wherein the rod is pre-formed with bends that match a profile of the vehicle.
11. A supplemental inflatable restraint system for a vehicle comprising: a first mount having a first barrel section having first tapered ends and having a first strap section that is fixed to the vehicle; a second mount having a second barrel section having second tapered ends and having a second strap section that is fixed to the vehicle; an inflatable bag; a retainer connected with the inflatable bag; and a rod extending between, and connected with, the first mount and the second mount, the rod extending through the retainer and configured to secure the inflatable bag to the vehicle, the first barrel section of the first mount and the second barrel section of the second mount each immovably fixed to the rod to maintain relative position, wherein the first tapered ends and the second tapered ends are disposed along and taper to the rod, wherein the first and second mounts are configured to retain the rod in a fixed position relative to the vehicle.
12. The supplemental inflatable restraint system of claim 11, wherein the retainer is configured to slip along the rod while being configured to remain secured to the rod during a deployment of the inflatable bag.
13. The supplemental inflatable restraint system of claim 11, wherein the rod is substantially rigid and conformable to a profile of the vehicle during installation of the inflatable bag in the vehicle.
14. The supplemental inflatable restraint system of claim 11, wherein the vehicle includes a body with a roof rail and an A-pillar, wherein the inflatable bag is secured to the roof rail and the A-pillar, wherein the rod is rigid and pre-formed before installation with bends that match a profile of the roof rail and the A-pillar.
15. The supplemental inflatable restraint system of claim 11, wherein the rod is one of a plural number of rods retaining the inflatable bag, wherein each of the rods is fixed to the vehicle by a pair of mounts.
16. The supplemental inflatable restraint system of claim 11, wherein the retainer comprises a sleeve through which the rod extends, wherein the sleeve extends substantially completely between the first mount and the second mount with a gap in the sleeve at the first mount.
17. The supplemental inflatable restraint system of claim 11, wherein the rod extends through the barrel section of each of the first and second mounts.
18. The supplemental inflatable restraint system of claim 11, wherein the rod extends through the barrel section of each of the first and second mounts, and wherein the strap section of each of the first and second mounts is secured to the vehicle by a fastener.
19. The supplemental inflatable restraint system of claim 11, wherein the rod extends from the second mount on both a first side of the second mount and a second side of the second mount, and wherein the inflatable bag is connected with the rod on both the first side and the second side.
20. A supplemental inflatable restraint system for a vehicle comprising: a body of the vehicle having a roof rail; a first mount having a first barrel section having first tapered ends and having a first strap section that is fixed to the roof rail; a second mount having a second barrel section having second tapered ends and having a second strap section that is fixed to the roof rail; an inflatable bag; a retainer connected with the inflatable bag, the retainer comprising a structure with loops, the structure including ends that are each fixed to the inflatable bag; and a rod extending between, and connected with, the first mount and the second mount, the rod extending through the loops of the retainer and configured to secure the inflatable bag to the vehicle, the first barrel section of the first mount and the second barrel section of the second mount each immovably fixed to the rod to maintain relative position, wherein the first tapered ends and the second tapered ends are disposed along and taper to the rod, wherein the first and second mounts are configured to retain the rod in a fixed position relative to the vehicle, wherein the rod is one of a plural number of rods retaining the inflatable bag, wherein each of the rods is fixed to the vehicle by a pair of mounts, wherein the vehicle includes the body with the roof rail, wherein the inflatable bag is

secured to the roof rail by the mounts and the rods, wherein the rods are rigid and pre-formed before installation with bends that match a profile of the roof rail.